

Selective Response for GenAI Under Forum Congestion

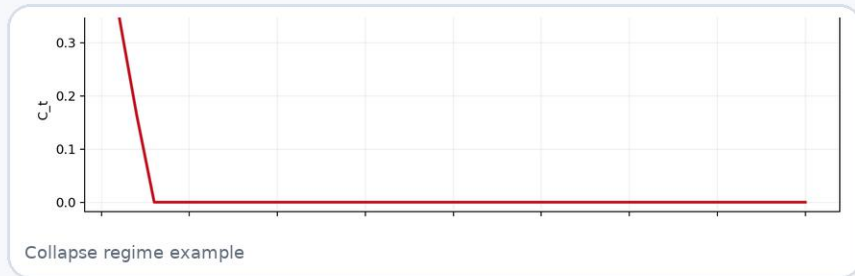
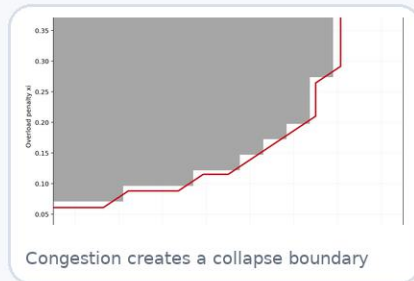
Game Theory 2026 mini project

Base paper: Taitler et al. (2025) | Our work: congestion + mechanism tests

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Talk in one line

- Selective response can help long-run learning, but forums are not infinite-capacity resources.
- We replicate the base paper's selective-response intuition and extend it with forum congestion and endogenous capacity.
- Main finding: novelty-aware routing outperforms uniform throttling, and the gain survives stricter controls.



Problem And Why It Matters

Routing users between GenAI and human forums changes both current service and future knowledge creation.

Core problem

- If GenAI answers everything, fewer high-value forum discussions happen, so future training signal shrinks.
- If GenAI defers too much, forum load rises, response quality drops, and expert capacity can collapse.

• The planner must manage both traffic

Course forums	Hard questions create reusable explanations for later students.
Developer communities	Stack Overflow style forums can be swamped by redirected users.
Product support	Short-term automation can reduce the human knowledge pipeline.



Congestion risk

- Load above capacity reduces useful throughput.
- Repeated overload damages future capacity.



Literature Review

Our project sits at the intersection of selective response, human-AI delegation, and congestion economics.

Selective response / reject option

Chow (1970), El-Yaniv (2010), Geifman and El-Yaniv (2017): when not to answer is part of the decision rule.

Learning to defer / human-AI systems

Madras et al. (2018), Mozannar and Sontag (2020), Kleinberg et al. (2018), Amershi et al. (2014): defer to humans when they add value.

Congestion / shared-resource view

Hardin (1968), Ostrom (1990), Kleinrock (1975), Vickrey (1969), Roughgarden and Tardos (2006): overloaded shared systems degrade.

Gap addressed by this project: prior selective-response work models knowledge creation, but not forum congestion and endogenous capacity loss.

This gap motivates our extension

Base Paper: Idea And Limits

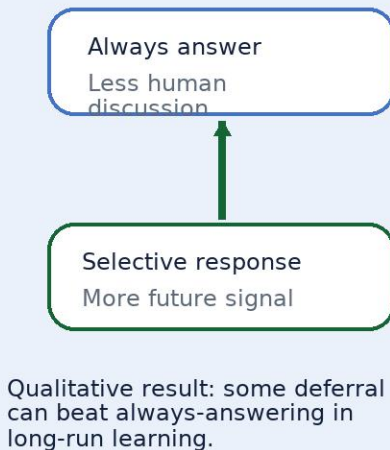
Selective-response intuition comes from the prior paper; congestion modeling does not.

Replication / prior work

What the base paper studies

- Question: should GenAI answer every query, or selectively defer some users to human communities?
- Key claim: answering less can improve long-run learning because deferred interactions create future human-generated knowledge.
- Interpretation: abstention is not only about current accuracy; it also shapes the future data ecosystem.

Base paper takeaway



Why that is not enough for our setting

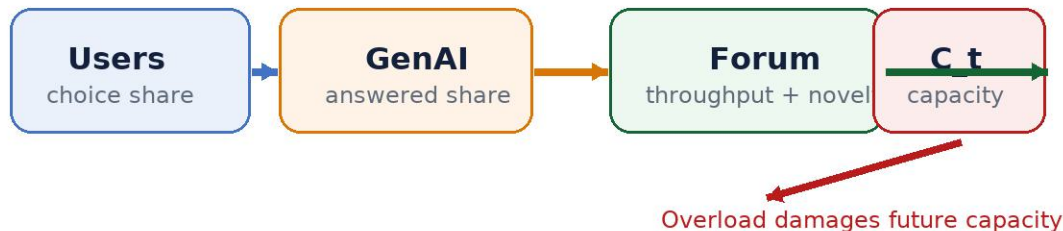
- Forum is treated as an implicit knowledge source, not a congestible shared resource.
- No endogenous forum capacity or collapse dynamics.
- No test of whether targeted gains survive when answered share is matched.
- No mechanism ablation to verify that novelty is the real source of the gain.

Our Original Work: Methodology

We extend the base idea with congestion dynamics, new baselines, and mechanism-validation experiments.

Our extension

Extended simulator



Key policies compared

- Uniform throttling: fixed answer rate.
- Targeted novelty policy: answer low-novelty items, defer high-novelty items.
- Capacity-aware baseline: answer more when recent overload is high.
- Novelty+capacity baseline: combine composition and congestion feedback.

Experiment design

40-round deterministic simulator
30 novelty seeds per comparison
3 regimes: stable, mid, collapse

New checks we added

- Matched answered-share control
- $\rho=0$ ablation to switch off novelty
- Collapse boundary in (C1, xi) plane

Main Results

Novelty-aware targeting beats uniform throttling across stable, mid, and collapse regimes.

Our result

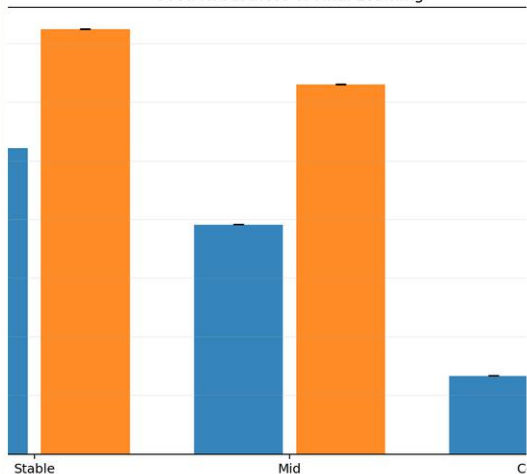
Headline numbers

- Stable regime: S_T rises from 26.15 to 36.29.
- Mid regime: S_T rises from 19.60 to 31.56 and overload drops from 40 to 25 rounds.

- Collapse regime: learning still improves from 6.68 to 9.54, but overload remains saturated.

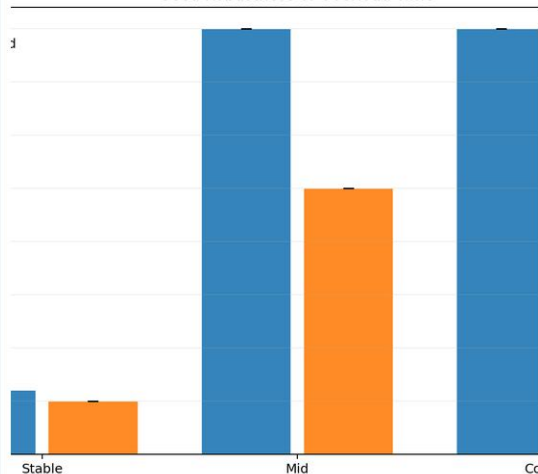
Targeting helps most when the forum is still productive enough for training purposes but is underper.

Seed Robustness of Final Learning



Final learning across 30 seeds

Seed Robustness of Overload Time



Overload-time comparison

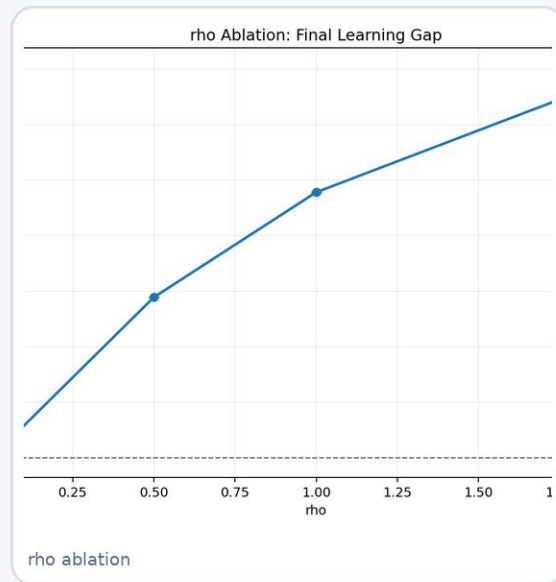
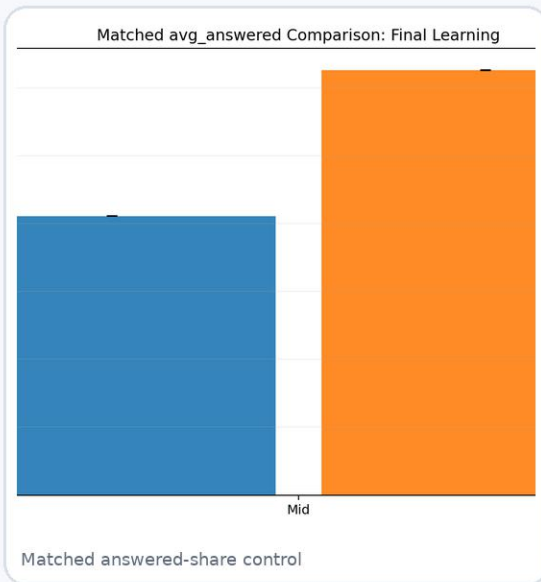
Mechanism Validation

We tested whether the targeted gain is real, not just an artifact of adoption or answer volume.

Our result

Two strongest checks

- Matched answered share: both policies answer about 0.178 of users, but targeted still reaches $S_T = 31.35$ vs 20.58 for uniform.
- rho ablation: when novelty has no effect ($\rho = 0$), the targeted advantage becomes exactly zero.
- Conclusion: the gain comes from sending better traffic to the forum, not just changing total answered volume.



Stronger Baselines And Tipping

Congestion feedback helps, but novelty+capacity performs best in the mid regime.

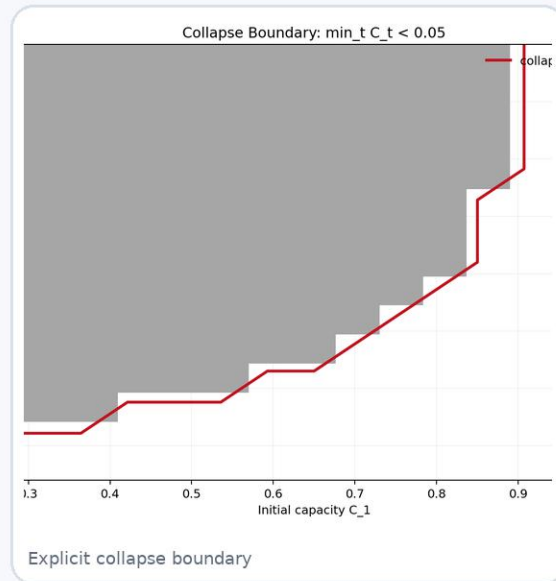
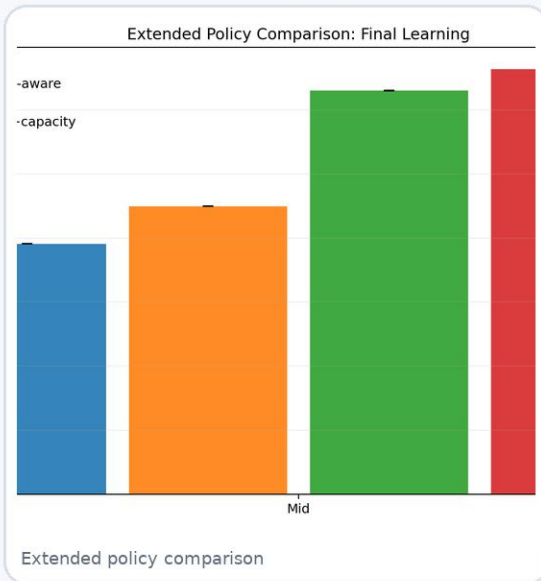
Our result

Mid-regime policy ranking

- Uniform: $S_T = 19.60$, overload = 40.
- Capacity-aware: $S_T = 22.52$, overload = 25.
- Targeted novelty: $S_T = 31.56$, overload = 25.
- Novelty+capacity: $S_T = 33.23$, overload = 18.

Phase transition

First collapse points appear near $C_1 \sim 0.364$ and $\xi \sim 0.074$. Higher overload damage requires higher initial forum capacity.



Summary And Future Work

References are listed on this slide; our contribution remains distinct from the base paper.

Summary

- Base paper insight: answering less can improve long-run learning.
- Our extension: forums are congestible, so routing must consider both quantity and quality of deferred traffic.
- Main result: novelty-aware routing dominates uniform throttling and survives strict controls.
- Best baseline in our mid regime combines novelty with congestion feedback.

Future work

- Estimate parameters from real forum data instead of reduced-form calibration.
- Replace hand-designed rules with learned routing policies.
- Model heterogeneous experts, delayed retraining, and multiple forums.
- Optimize jointly for learning, welfare, and revenue instead of one metric.

References used in the talk

- Taitler et al. (2025)
- Chow (1970)
- El-Yaniv (2010)
- Madras et al. (2018)
- Mozannar and Sontag (2020)
- Hardin (1968)
- Ostrom (1990)
- Vickrey (1969)
- Kleinrock (1975)

Practice target for 8 minutes: ~45-55 seconds per slide. Keep slide 4 explicitly labeled as prior work and slides 5-8 as our contribution.